School of Computer Science and Engineering

Fall Semester 2026-27

BCSE331P – Exploratory Data Analysis

Assessment 1: Merging and Data Transforamtion CO1 BL2

CO1: Engrave simple R programs.

Dataframe Operations

Write an R program to create three data frames named **dataframe1**, **dataframe2**, and **dataframe3**.

- **dataframe1** should contain the attributes: **ProductID**, **NameofProd**, and **Price**.
- **dataframe2** should contain the attributes: **ProductID** and **Company_Location**.
- **dataframe3** should contain the attributes: **ProductID**, **NameofProd**, and **Price**.

Insert a minimum of **10 records** into each data frame.

Using the above data frames, perform the following operations:

1. Append the records of **dataframe1** and **dataframe3**.
2. Concatenate the data frames wherever applicable.
3. Merge **dataframe1** and **dataframe2** using the **ProductID** attribute.
4. Perform the following join operations between **dataframe1** and **dataframe2**:
    - Inner Join
    - Left Join
    - Right Join
    - Full Outer Join

Data Transformations

Write an R program to create a numeric vector using your **first name (Fname)** as the object name. The vector should contain **10 numeric values**. The first value must be the **last four digits of your Registration Number**, followed by **any nine integer values** of your choice.

Perform the following data transformation techniques on the vector:

1. Normalize the data to the following ranges:
    - `[-3, +3]`
    - `[-1, +1]`
    - `[0, 1]` (Min-Max Normalization)
2. Transform the data into **Z-score** format.
3. Transform the data using **Decimal Scaling Normalization**.